THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — MATHS TWO-CHAPTER TEST SET B

Chapters 1 & 2: **Orienting Yourself: The Use of Coordinates, Introduction to Linear Polynomials** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat figures wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The distance of the point $(-5, 12)$ from the origin is: 1

(a) 13 units **(b)** 17 units **(c)** 14 units **(d)** 5 units

2. The reflection of the point $(-2, 6)$ in the y-axis is: 1

(a) $(6, -2)$ **(b)** $(2, 6)$ **(c)** $(-2, -6)$ **(d)** $(2, -6)$

3. A point lies on the x-axis. Its y-coordinate is: 1

(a) 1 **(b)** 0

(c) negative **(d)** equal to its x-coordinate

4. Which of the following points does not lie in Quadrant II? 1

(a) $(-4, 4)$ **(b)** $(-1, 9)$ **(c)** $(4, -4)$ **(d)** $(-7, 2)$

5. The distance of the point $(-7, 4)$ from the x-axis is: 1

(a) 7 units **(b)** 4 units **(c)** −4 units **(d)** 11 units

6. For the line $y = 4x - 7$, the slope and the y-intercept are respectively: 1

(a) -7 and 4 **(b)** -4 and -7

(c) 4 and -7 **(d)** 4 and 7

7. The point $(7, 15)$ ___ the line $y = 2x + 1$. 1

(a) does not lie on the line **(b)** is the origin

(c) is the y-intercept of the line **(d)** lies on the line

8. Which of the following is a linear polynomial? 1

(a) $5x - 8$ **(b)** $x^2 + 1$ **(c)** $x^3 - x$ **(d)** 7

9. The lines $y = 3x + 2$ and $y = 3x - 5$ are: 1

(a) perpendicular **(b)** intersecting at the origin

(c) coincident **(d)** parallel

10. A student claims: "$5x^2 - 3$ is a linear polynomial." The correct response is: 1

(a) True only for $x > 0$

(b) False, its degree is 2 so it is quadratic

(c) True

(d) Cannot be determined

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Show that the points $A(1, 1)$, $B(3, 3)$ and $C(5, 5)$ are collinear using distances. 2

12. The midpoint of AB is $M(2,\ -1)$ and $A = (-4,\ 3)$. Find B. 2

13. State the slope and y-intercept of $y = -4x + 9$ and say whether it shows linear growth or decay. 2

14. The height of water in a tank is $h(t) = 3 - 0.5t$ metres after t months. Find the height after 5 months and the time when the tank is empty. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Show that the point $(0,\ -1)$ is equidistant from $(-4,\ 3)$ and $(4,\ 3)$. Hence name the type of triangle formed by the three points, with reason. 3

16. Write the coordinates of three points on the line $y = 3x - 2$ and state where it cuts the y-axis. 3

17. The points $(1,\ -8)$, $(-4,\ 7)$ and $(-7,\ -4)$ lie on a circle with centre at the origin. Find its radius and say whether $(-5,\ 6)$ lies inside, on or outside the circle. 3

OR

Find the coordinates of the fourth vertex of a parallelogram $ABCD$ with $A(1,\ 2)$, $B(4,\ 3)$, $C(6,\ 6)$. 3

18. The points $(-3, 6)$, $(-2, 4)$, $(0, 0)$, $(1, -2)$ lie on a line. Find its equation and its slope. What does the slope tell? 3

OR

The area of a rectangle of length 13 cm and breadth b cm is a linear polynomial in b. Find it for $b = 12, 10, 8$ and write the pattern. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Show that $(-4,\ -7)$, $(-1,\ 2)$, $(8,\ 5)$ and $(5,\ -4)$ are the vertices of a rhombus. Find its area using the diagonals. 5

OR

Find the point on the x-axis which is equidistant from $(-2,\ 5)$ and $(2,\ -3)$, and find the common distance. 5

20. The points $(0,\ 5)$ and $(4,\ 13)$ lie on a line $y = ax + b$. 5

(i) Find a and b.
(ii) Write the equation.
(iii) Is $(10, 25)$ on the line?
(iv) Find the point where the line cuts the x-axis.

OR

Case study: Reiaan's school sells notebooks at ₹x each. The cost for n notebooks including a one-time ₹20 bag fee is $20 + x\,n$. For 5 notebooks the bill is ₹70. 5

(i) Find x.
(ii) Write the cost function.
(iii) Find the cost of 12 notebooks.
(iv) How many notebooks can be bought for ₹170?

END OF QUESTION PAPER — Total: 40 Marks